

A Comprehensive Measure of Well-Being

Abstract:

The Human Development Index (HDI) is an internationally recognized indicator developed by the United Nations Development Programme (UNDP) to measure the overall development of a country based on health, education, and standard of living. This project presents a Machine Learning-based web application that predicts the HDI score using four important socio-economic indicators: Life Expectancy at Birth, Expected Years of Schooling, Mean Years of Schooling, and Gross National Income (GNI) per Capita.

The application is developed using Python, Flask, HTML, CSS, and Scikit-learn. A Linear Regression model is trained on the HDI dataset and integrated into a Flask web application to provide real-time predictions. The application offers a simple and user-friendly interface that allows users to enter the required values and instantly view the predicted HDI score along with its corresponding development category.

Problem Statement:

Evaluating a country's overall development involves analyzing multiple socio-economic factors such as health, education, and income. Traditional methods of calculating the Human Development Index require extensive data processing and statistical analysis, making it difficult to obtain quick predictions.

The objective of this project is to develop a Machine Learning-based web application that predicts the Human Development Index (HDI) using key development indicators. The system provides fast and accurate predictions through a simple web interface, helping users understand a country's level of human development.

Features:

- Predicts Human Development Index (HDI) using Machine Learning.
- User-friendly web interface built with Flask.
- Country selection using a dropdown menu.
- Accepts user input for:
 - Life Expectancy at Birth
 - Expected Years of Schooling
 - Mean Years of Schooling
 - Gross National Income (GNI) per Capita
- Generates real-time HDI predictions.
- Displays HDI category:
 - Low Human Development
 - Medium Human Development

- High Human Development
- Very High Human Development
- Responsive web interface using HTML and CSS.
- Pre-trained Linear Regression model for faster predictions.

Technology Stack:

Component	Technology Used
Programming Language	Python 3.11
Machine Learning	Scikit-learn
Data Processing	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn
Web Framework	Flask
Frontend	HTML5, CSS3
IDE	Visual Studio Code, Jupyter Notebook
Version Control	Git & GitHub

Installation Steps:

Clone the GitHub repository.

```
git clone https://github.com/Marenhemalatha/Comprehensive-measure-of-well-being.git
```

Step 2

Navigate to the Flask folder.

```
cd Flask
```

Step 3

Install the required dependencies.

```
pip install -r requirements.txt
```

Step 4

Run the Flask application.

```
python app.py
```

Step 5

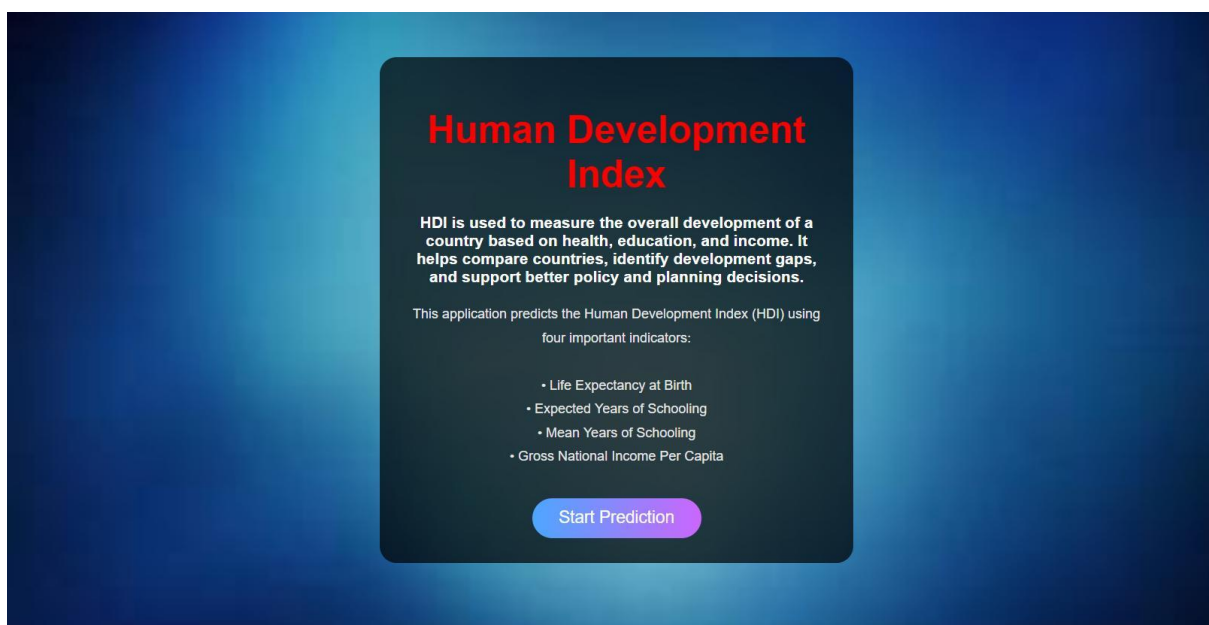
Open your browser and visit:

<http://127.0.0.1:5000>

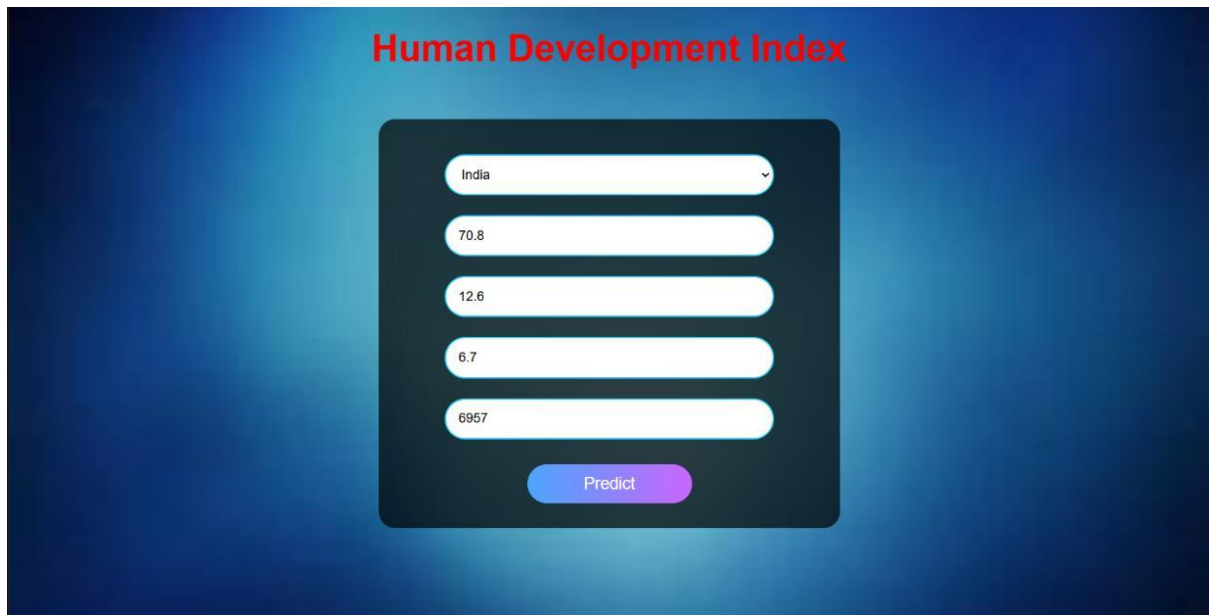
Usage Instructions

1. Open the application in your web browser.
 2. Click the Start Prediction button.
 3. Select a country from the dropdown list.
 4. Enter:
 - Life Expectancy at Birth
 - Expected Years of Schooling
 - Mean Years of Schooling
 - Gross National Income (GNI) per Capita
 5. Click the Predict button.
 6. View the predicted Human Development Index score and its corresponding development category.
 7. Click Predict Again to make another prediction.
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Screenshots:



Human Development Index



A screenshot of a web application interface for predicting the Human Development Index (HDI) for India. The interface is set against a dark blue gradient background. It features a central dark blue rounded rectangle containing five white input fields and a 'Predict' button. The input fields are pre-filled with the following values: 'India' (in a dropdown menu), '70.8', '12.6', '6.7', and '6957'. The 'Predict' button is a rounded rectangle with a blue-to-purple gradient.

Country	GDP per capita	Life expectancy	Mean years of schooling	Expected years of schooling
India	70.8	12.6	6.7	6957

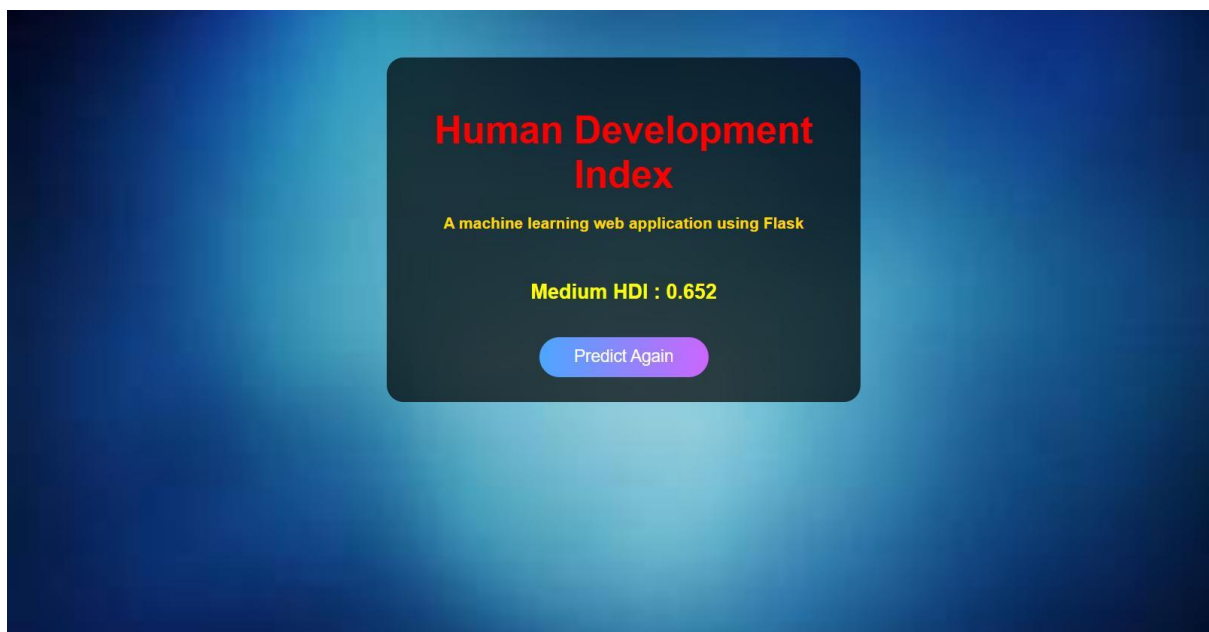
Predict

Human Development Index

A machine learning web application using Flask

Medium HDI : 0.652

Predict Again



A screenshot of the result page of the Human Development Index prediction application. It features a central dark blue rounded rectangle on a dark blue gradient background. The rectangle contains the title 'Human Development Index' in red, a subtitle 'A machine learning web application using Flask' in yellow, the result 'Medium HDI : 0.652' in yellow, and a 'Predict Again' button with a blue-to-purple gradient.

Future Scope:

- Integrate the latest HDI datasets for improved prediction accuracy.
- Implement advanced Machine Learning algorithms such as Random Forest and XGBoost.
- Automatically populate country-specific values from the dataset.
- Add interactive charts and data visualizations.
- Deploy the application on cloud platforms such as Render or AWS.
- Develop a mobile-friendly version of the application.
- Enhance input validation and error handling.
- Include additional socio-economic indicators for more accurate predictions.

Team Member Details:

S.No	Team Member	Role
1	Marem Hemalatha	Documentation Preparation, Application Demonstration, Code Explanation, GitHub Repository Management
2	Jarapala Padma Bai	Dataset Collection, Data Preprocessing, Machine Learning Model Development
3	Prasanna Kogila	Flask Application Development, Frontend Integration (HTML & CSS)
4	Sailekhana Thuraka	Testing, Validation, Project Integration, Documentation Review

Project Structure

Comprehensive-measure-of-well-being

```
|
|  └─ Dataset
|  |   └─ HDI.csv
|  |
|  |  └─ Flask
|  |  |  └─ app.py
|  |  |  └─ HDI.pkl
|  |  |  └─ requirements.txt
|  |  |  └─ static
|  |  |  |  └─ css
|  |  |  |  |  └─ style.css
|  |  |  |  └─ images
|  |  |  |  |  └─ world.jpg
|  |  └─ templates
|  |  |  └─ home.html
|  |  |  └─ indexnew.html
|  |  |  └─ result.html
|  |
|  |  └─ Training
|  |  |  └─ HumDevIndex.ipynb
|  |
|  └─ README.md
```

Conclusion

The Human Development Index Prediction System successfully demonstrates the application of Machine Learning in predicting a country's level of development based on key socio-economic indicators. By integrating a Linear Regression model with a Flask web application, the project provides an efficient, accurate, and user-friendly solution for real-time HDI prediction. The project highlights the practical use of data science and web technologies in supporting development analysis and serves as a foundation for future enhancements.